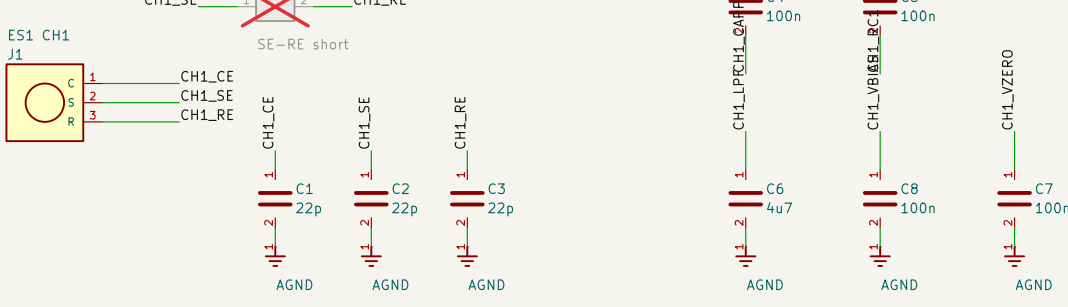


4-CHANNEL EC SENSE ES1 ELECTROCHEMICAL GAS SENSOR FRONT END – SINGLE I2C BUS OUTPUT

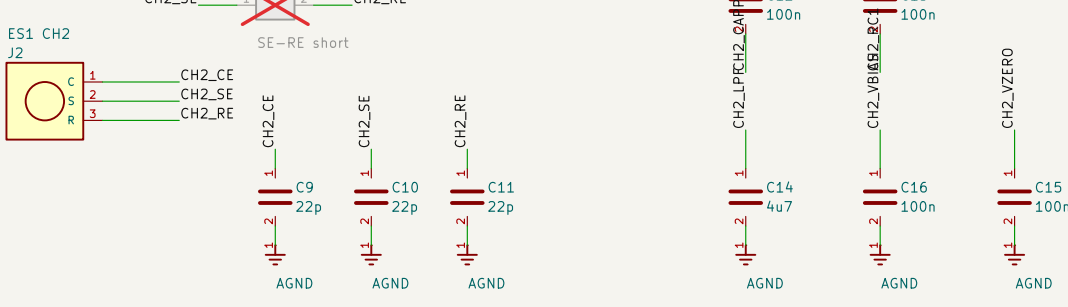
2 x ADuCM355 dual potentiostat, each an I2C target (U1=0x30, U2=0x31). Programmable bias per channel.

- REV B – PCB / ASSEMBLY NOTES
1. Pinout and decoupling corrected to ADuCM355 Rev C. Board: 120 x 80 mm, 4 layers.
  2. All capacitors fitted. JP1–JP4 stay OPEN during operation: R11 (0 ohm) must be fitted.
  3. Sensor sockets: 3 x Mini-Max (S948–0–15–15–21–27–10–0 per ES1, 1.0 mm finished PTH).
  4. PCB top-view electrode order follows the mirrored manufacturer bottom-view drawing.
  5. U1/U2 require Firmware. Configure 0x30/0x31 from address straps: ALERT must be open drain.
  6. Perform clean-air zero, certified-gas span calibration and temperature compensation. This hardware prototype has not been bench-tested. See PCB\_README.md.

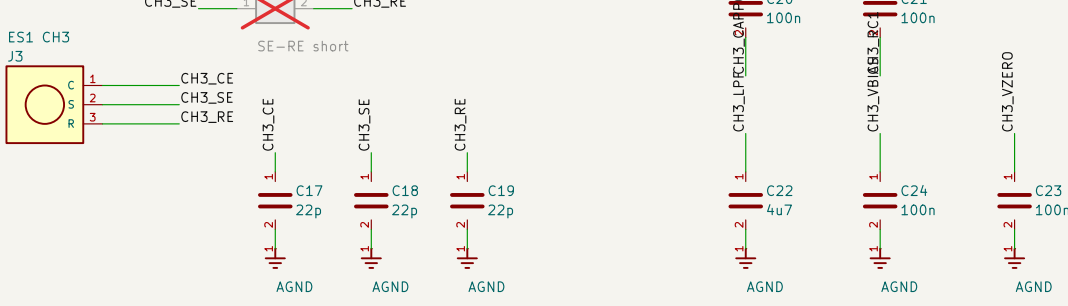
CHANNEL 1 (U1 potentiostat 0)



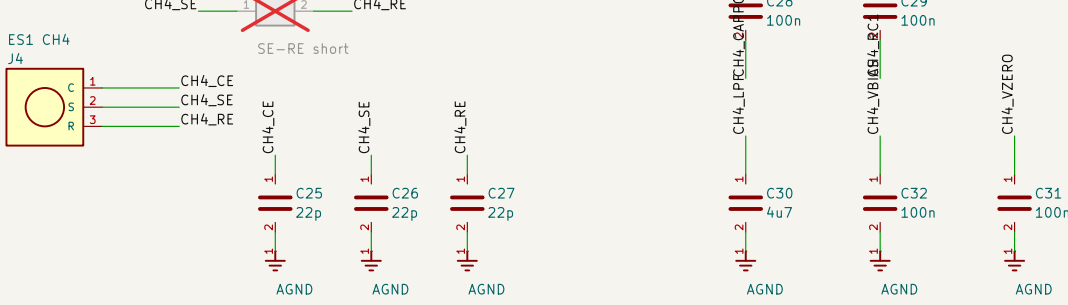
CHANNEL 2 (U1 potentiostat 1)



CHANNEL 3 (U2 potentiostat 0)



CHANNEL 4 (U2 potentiostat 1)



DESIGN NOTES

RTIA selection (ADuCM355 LPTIA, programmable 1k–512k):

ES1–AG1–200 : 40 nA/ppm, 200 ppm FS => 8 uA FS, RTIA = 100k => 0.80 V FS.

16-bit ADC over +/-0.9 V => -27 uV LSB => 0.27 nA => -0.007 ppm/LSB.

ES1–AG1–10 : 40 nA/ppm, 10 ppm FS => 400 nA FS, RTIA = 512k => 0.205 V FS => -0.0013 ppm/LSB.

Keep RLLOAD = 100 R (EC Sense recommended load resistor for the ES1 series).

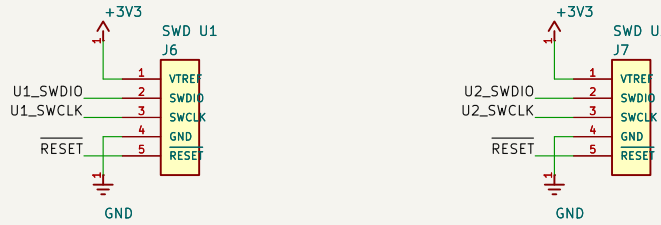
Bias: VBIAS/VZERO 12-bit LPDACs, 0.2–2.4 V, 0 mV bias => VBIAS = VZERO (typ. 1.1 V).

For biased gases set VBIAS = VZERO + Vbias\_target.

I2C: each ADuCM355 runs as an I2C target. Firmware reads its strap on P2.4 at boot and selects 0x30 or 0x31, so one binary serves both devices.

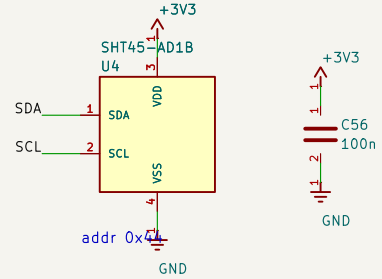
Layout: guard ring at VZERO potential around every SE net; no soldermask opening under the sensor sockets; clean with no-clean flux only; keep SE traces < 15 mm; AGND and DGND joined at one point under U1/U2; do not route I2C or SWD under the AFE section.

DEBUG / PROGRAMMING

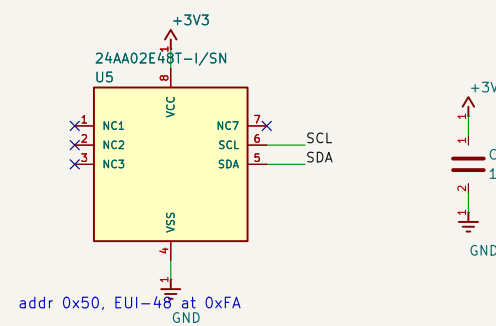


U1 I2C target 0x30

AMBIENT T / RH (mandatory for compensation)

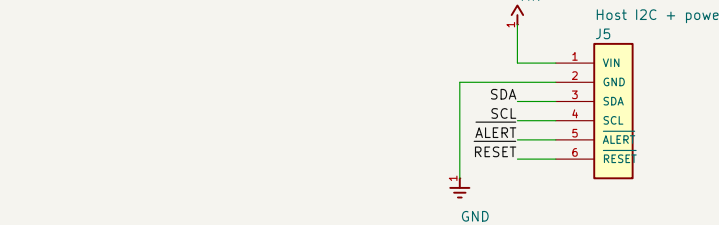


BOARD ID / CAL STORE

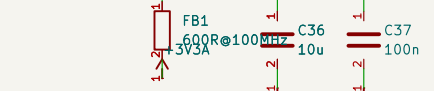


U2 I2C target 0x31

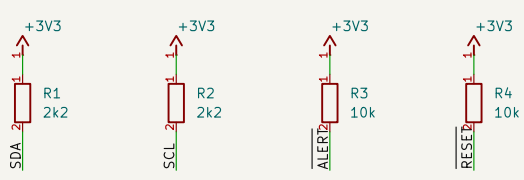
POWER



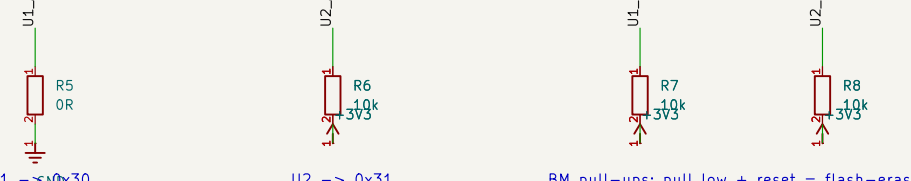
Analog rails: ferrite bead isolated from +3V3



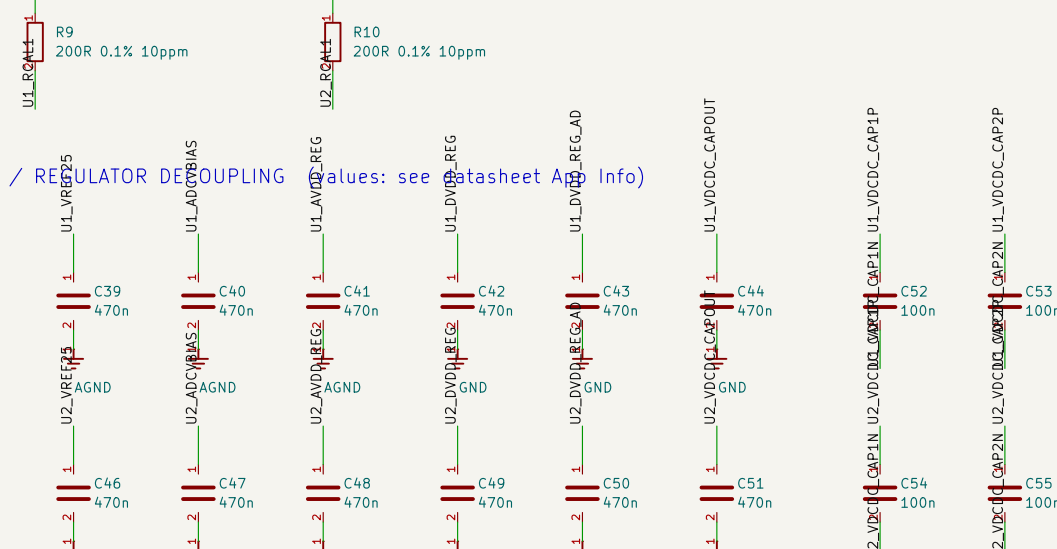
I2C BUS



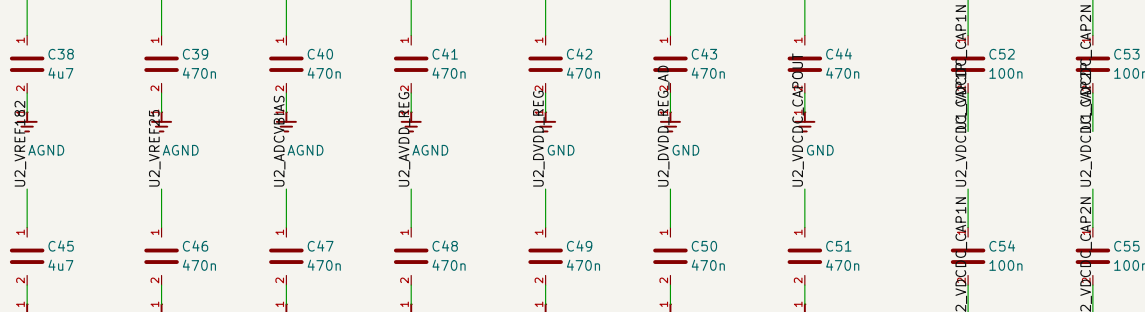
I2C ADDRESS STRAPS (read on P2.4 at boot)



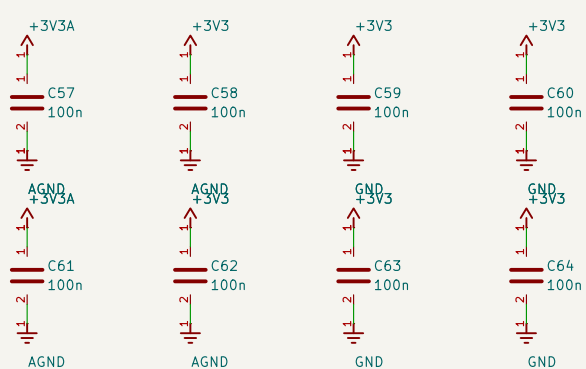
EIS CALIBRATION RESISTORS



REFERENCE / REGULATOR DECOUPLING (values: see datasheet for info)



LOCAL SUPPLY BYPASS – PLACE AT EACH IC POWER PIN



AGND–GND TIE + POWER FLAGS

